

RK Series

Microminiature SMT Top Actuated Tactile Switch



Description

RK series is low profile micro miniature SMT top actuated tactile switch with different choices of actuation force and long life cycle. The switch is ideal for applications that need a high density of components in limited PCB space, such as automotive key fob, navigation and health & medical devices. The compact top-actuated RK series comes in various product sizes (RKA2: 4.5 x 4.5 x 0.55 mm, RKB2: 4.2 x 3.2 x 2.5 mm, RKC: 3.5 x 2.6 x 1.4 mm). The RK series is available in 6 actuation forces, allowing customers easily integrate into different designs.

Features & Benefits

- Low profile & microminiature package
- Hard actuator
- SMT termination
- Different actuation forces
- Life cycles up to 1 Mio

Specifications

Function	Momentary action
Contact Arrangement	1 make contact = SPST N.O.
Terminals	SMT termination

Electrical Characteristics

Dielectric Strength	250 VA C (1 minute)
Bounce Time	≤ 10 ms

Type	Max. Voltage	Max. Current	Contact Resistance	Insulation Resistance
RKA2	DC 15 V	20 mA max.	500mΩ max.	50MΩ min.
RKB2	DC 12V	50 mA max.	200mΩ max.	100MΩ min.
RKC	DC 12V	50 mA max.	2Ω max.	100MΩ min.

Environmental Characteristics

Operating Temperature	RKA2: -40°C to 85°C RKB2: -40°C to 90°C RKC: -40°C to 85°C
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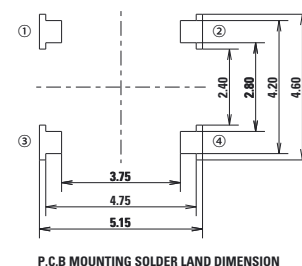
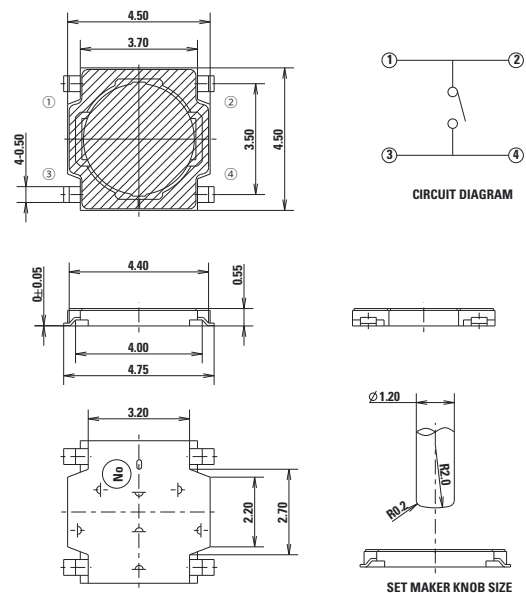
Process

Soldering	This component is suited to the following methods: Infrared Reflow Soldering in accordance with IEC61760-1
Packaging	RKA2: 5,000 pieces per reel RKB2: 3,000 pieces per reel RKC: 9,500 pieces per reel

Applications

- Keyfob
- Health & medical devices
- Navigation

RKA2 4.5 x 4.5 x 0.55 (mm)

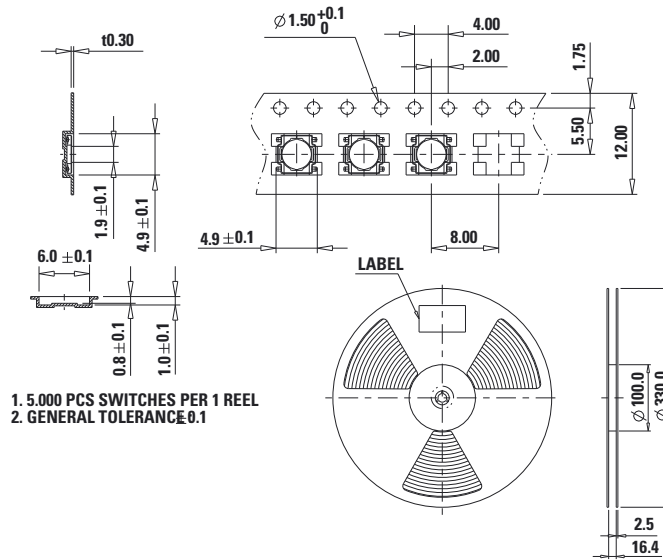


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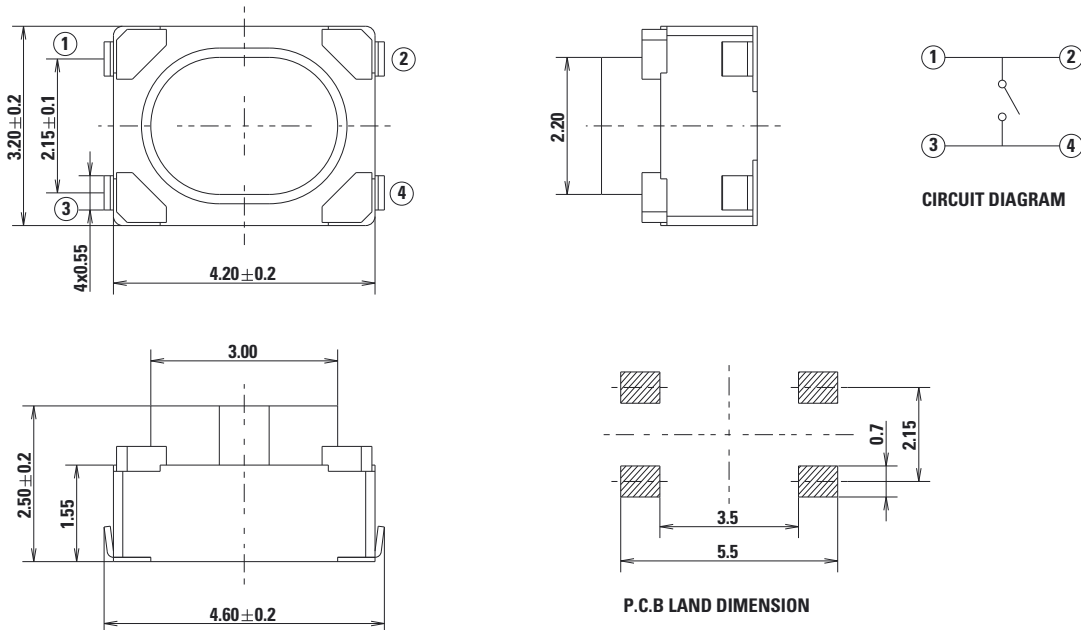
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Tape & Reel



RKB2 4.2 x 3.2 x 2.5 (mm)

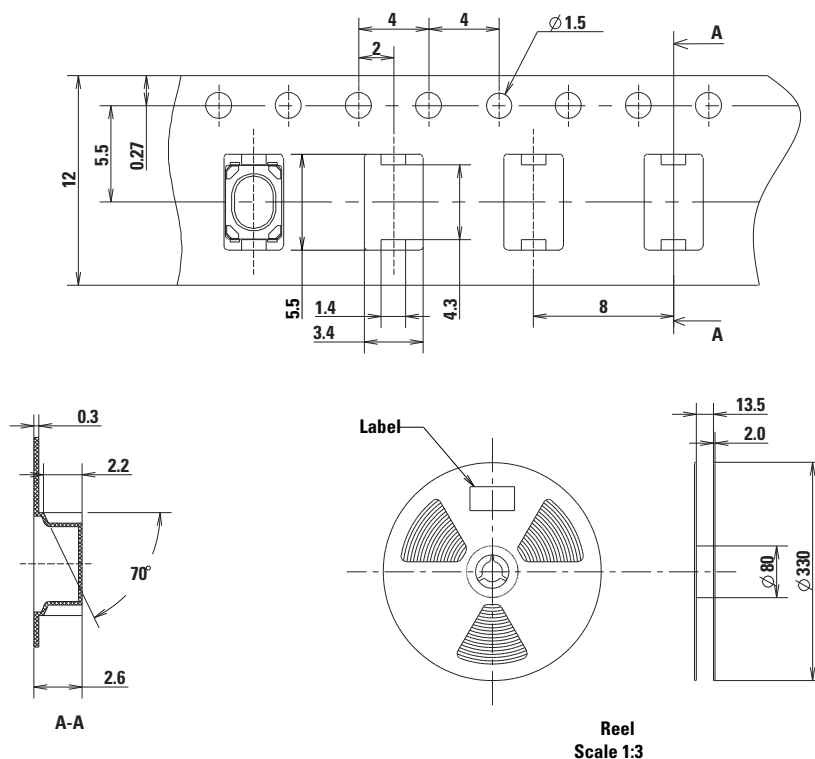


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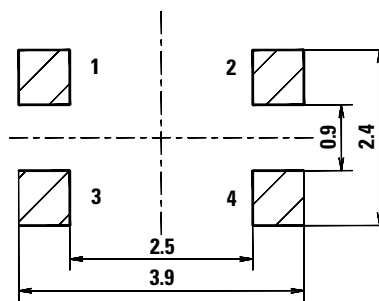
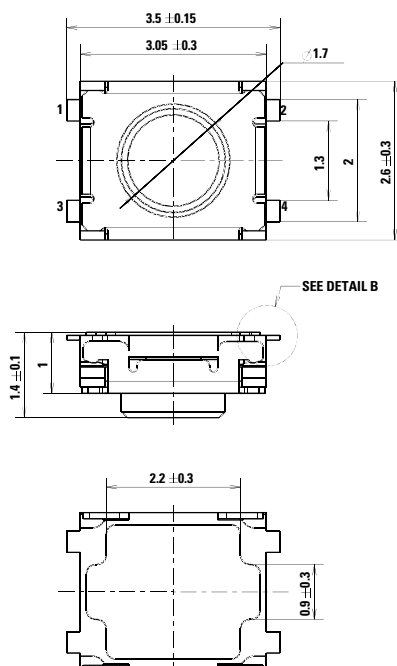
Tape & Reel



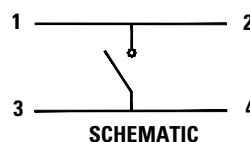
Notes:

1. 3K pcs Switches per reel.
2. General tolerance: 0.3mm

RKC 3.5 x 2.6 x 1.4 mm



RECOMMENDED PCB LAYOUT

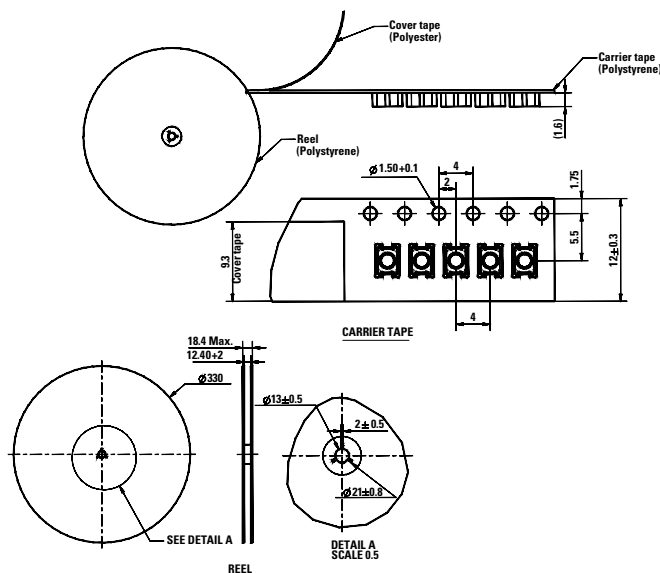


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Tape & Reel



Ordering Numbers

Our easy build-a-switch concept allows you to mix and match options to create the switch you need. To order, select desired option from each category and place it in the appropriate box. For any part number different from those listed above, please consult your local representative.

RKA2 - S - G - x - 055 - SMTR - LFS

Termination:
G: Gullwing
Actuation Force
N: 1.0N
M: 1.6N
K: 2.5N
G: 4.0N
S: 5.4N
Height
055: 0.55 mm Height
LFS: Lead free compatible

RKB2 - S - J - x - 250 - SMTR - LFS

Termination:
J: J Bend SMT
Actuation Force
M: 1.8N
K: 2.5N
G: 4.0N
Height
250: 2.5 mm Height
LFS: Lead free compatible

RKC - G - G - 140 - SMTR - x - LFS

Termination:
G: Gullwing
Actuation Force
G: 4.5N
Height
140: 1.4 mm
LFS: Lead free compatible
Grounding Option
G: Ground Pin
None: Without Ground Pin

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Mechanical Characteristics

Supply Status	Littelfuse Orderable Part Number	Description	Contact & Terminals Plating	Terminals	Ground Pin	Operating Force Fa (Newton)	Operating Life (Operations)	Electrical Travel (mm)	Product Height from PCB (mm)
✓	10913237	RKA2SGG055SMTR LFS	Silver	Gull Wing SMT	No	4.0 +/- 0.7	200,000	0.20 ± 0.1	0.55
✓	10913231	RKA2SGK055SMTR LFS	Silver	Gull Wing SMT	No	2.5 +/- 0.5	200,000	0.20 ± 0.1	0.55
✓	10913230	RKA2SGM055SMTR LFS	Silver	Gull Wing SMT	No	1.6 +/- 0.5	1,000,000	0.20 ± 0.1	0.55
✓	G01138607	RKA2SGN055SMTR LFS	Silver	Gull Wing SMT	No	1.0 +/- 0.5	1,000,000	0.20 ± 0.1	0.55
✓	10913238	RKA2SGS055SMTR LFS	Silver	Gull Wing SMT	No	5.4 +/- 0.7	150,000	0.20 ± 0.1	0.55
✓	10896914	RKB2SJG250SMTR LFS	Silver	J-lead SMT	No	4.0 +/- 0.5	100,000	0.20 ± 0.1	2.5
✓	10890919	RKB2SJK250SMTR LFS	Silver	J-lead SMT	No	2.5 +/- 0.5	100,000	0.20 ± 0.1	2.5
✓	10890915	RKB2SJM250SMTR LFS	Silver	J-lead SMT	No	1.8 +/- 0.5	100,000	0.20 ± 0.1	2.5
✓	Y94ST24B8QBFP	RKC GG140 G SMTR LFS	Silver	Gull Wing SMT	Yes	4.5 +/- 0.8	200,000	0.15 ± 0.1	1.4
✓	Y94ST23B8QBFP	RKC GG140 SMTR LFS	Silver	Gull Wing SMT	No	4.5 +/- 0.8	200,000	0.15 ± 0.1	1.4

Notes: Supply status: ✓ : Available for ordering;

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